Change of variables

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Stone-Weierstrass theorem

Let $\mathcal{A}$ be an algebra of real continuous functions on a compact set $K\subset \R^n$. If $\mathcal{A}$ separates points on K and if $\mathcal{A}$ vanishes at no point of K, then the uniform closure $\mathcal{B}$ of $\mathcal{A}$ consists of all real continuous functions on K.

Integration

Suppose that I^k is a k-cell in $\R^k$ consisting of all $$ \x=(x_1,\cdots,x_k) $$ such that

$$a_i \leq x_i \leq b_i \qquad i = 1, ..., k$$

I^j is the j-cell in $\R^j$ defined by the first j inequalities, and f is a real continuous function on I^k.

Put $f = f_k$ and define f_{k-1} on I^{k-1} by $$ f_{k-1}(x_1,\cdots,x_{k-1})=\int _{a_k} ^{b_k} f_k(x_1, \cdots,x_{k-1},x_k)\d x_k $$ The uniform continuity of f_k on I^k shows that f_{k-1} is continuous on I^{k-1}. Hence we can repeat this process and obtain functions f_j, contiunous on I^j, such that f_{j-1} is the integral of f_j with respect to x_j over $\left[a_j, b_j\right]$. After k steps we arrive at a number f_0, which we call the *integral of f over I^k*; we write it in the form $$ \int_{I^k} f(\x)\d\x \,. $$

A priori, this definition of the integral depends on the order in which the k integrations are carried out. To prove thiss, let us introduce the temporary notation $L(f)$ for the integral and $L'(f)$ for the result obtained by carring out the k integrations in some other order.

For every $f \in C\left(I^k\right)$, $L(f) = L'(f)$.

Proof. If $h(\x)=h_1(x_1)\cdots h_k(x_k)$, where $h_j \in C\left[a_j, b_j\right]$, then $$ L(h)=\prod_{i=1}^k \int _{a_i}^{b_i}h_i(x_i)\d x_i=L^{\prime}(h). $$ If $\mathcal{A}$ is the set of all finite sums of such functions h, it follows that $L(g) = L'(g)$ for all $g \in \mathcal{A}$. Also, $\mathcal{A}$ is an algebra of functions on I^k to which the *Stone-Weierstrass Theorem* applies.

Put $V = \prod_{i=1}^k (b_i - a_i)$. If $f \in C\left(I^k\right)$ and $\epsilon > 0$, there exists $g \in \mathcal{A}$ such that $\parallel f - g \parallel < \epsilon/V$, where $\parallel f \parallel$ is defined as $\max \vert f(\x) \vert\,(\x \in I^k)$. Then $\mid L(f - g) \mid < \epsilon$, $\mid L'(f - g) \mid < \epsilon$, and since

$$L(f) - L'(f) = L(f - g) + L'(g - f)$$

we conclude that $| L(f) - L'(f) | < 2\epsilon$.

The *support* of a (real or complex) function f on $\R^k$ is the closure of the set of all points $\x \in \R^k$ at which $f(\x)\neq 0$. If f is a continuous function with compact support, let I^k be any k-cell which contains the support of f and define $$ \int _{\R^k}f:=\int_{I^k}f\,. $$ The integral so defined is evidently independent of the choice of I^k, provided only that I^k ccontains the support of f.

Primitive mappings

If $\mathbf{G}$ maps an open set $E \subset \mathbf{R}^n$ into $\mathbf{R}^n$, and if there is an integer m and a real function g with domain E such that

$$\begin{equation} \mathbf{G}(\x)= \sum_{i \neq m} x_i \mathbf{e}_i+g(\x) \mathbf{e}_m, \qquad \x \in E \end{equation} \tag{1}$$

then we call $\mathbf{G}$ *primitive*.

A primitive mapping is thus one that changes at most one coordinate; If g is differentiable at some point $\aa\in E$, so is $\G$. The matrix $\left[\alpha_{ij}\right]$ of the operator $(\d\G)_{\aa}$ has $$ [(\frac{\partial g}{\partial x_1})_{\aa},\cdots,(\frac{\partial g}{\partial x_m})_{\aa},\cdots,(\frac{\partial g}{\partial x_n})_{\aa}] $$ as its m'th row. For $j \neq m$, we have $\alpha_{jj} = 1$ and $\alpha_{ji} = 0$ if $i \neq j$. The Jacobian of $\G$ at $\aa$ is thus given by $$ J_G(\aa)=\det[(\d \G)_{\aa}]=(\frac{\partial g}{\partial x_m})_{\aa}\,. $$ Clearly $(\d\G)_{\aa}$ is invertible if and only if $(\frac{\partial g}{\partial x_m})_{\aa}\neq 0$.

(a) A linear operator B on $\R^n$ that interchanges some pair members of the standard basis and leaves the others fixed is called a *flip*.

(b) By $P_m, 0 \leq m \leq n$ we mean the projections in $\R^n$: $P_0 \x=0$ and $$ P_m \x=x_1\mathbf{e}_1+\cdots+x_m\mathbf{e}_m,\qquad x\in\R^n\,. $$

Theorem of local representation

Suppose $\mathbf{F}$ is a C^1-mapping of an open set $E\subset \R^n$ into $\R^n$, $\mathbf{0} \in E$, $\mathbf{F}(\mathbf{0}) = \mathbf{0}$, and $(\d\mathbf{F})_{\mathbf{0}}$ is invertible. Then there is a neighbourhood of $\mathbf{0}$ in $\R^n$ in which a representation

$$\begin{equation} \mathbf{F}(\x)=B_1 \circ \cdots \circ B_{n-1} \circ \G_n \circ \cdots \circ \mathbf{G}_1(\x) \end{equation} \tag{2}$$

is valid.

In Equation 2, each $\G_i$ is a primitive C^1-mapping in some neighborhood of $\mathbf{0}$; $\G_i(\mathbf{0})=\mathbf{0}$, $(\d\G_i)_0$ is invertible, and each B_i is either a flip or the identity operator.

Briefly, Equation 2 represents $\mathbf{F}$ locally as a composition of primitive mappings and flips.

Proof. Assume that for each $1 \leq m \leq n-1$ there exists a neighborhood V_m of $\0$ and a C^1-mapping $\F_m$ on it, with $\F_m(\0)=\0$ and $(\d\F_m)_0$ invertible, for which the following holds:

$$P_{m-1} \circ F_m(x)=P_{m-1} x \qquad (x \in V_m). \tag{3}$$

By Equation 3, we have $$ \F_m(\x)=P_{m-1}\x+\sum_{i=m}^{n}\alpha_i(\x)\mathbf{e}_i, $$ where $\alpha_m, \ldots, \alpha_n$ are real C^1-functions in V_m. Hence

$$(\d F_m)_0 \, , \mathbf{e}_m= \sum_{i=m}^{n}\left(\frac{\partial \alpha_i}{\partial x_m}\right)_0 \mathbf{e}_i \tag{4}$$

Since $(\d\F_m)_\0$ is invertible, the left side of Equation 4 is not $\0$, and therefore there is a k such that $m \leq k \leq n$ and $(\frac{\partial \alpha_k}{\partial x_m})_\0\neq 0$.

Let B_m be the flip that interchanges this m and this k (if $k = m$, B_m is the identity) and define $$ \G_m(\x)=\x+[\alpha_k(\x)-x_m]\mathbf{e}_m \qquad \x\in V_m\,. $$ Then $\G_m\in C^1(V_m)$, $\G_m$ is primitive and $(\d\G_m)_\0$ is invertible.

The inverse function theorem shows therefore that there is an open set U_m with $\0\in U_m\subset V_m$ such that $\G_m$ is a 1-1 mapping of U_m onto a neighborhood V_{m+1} of $\0$, in which $\G^{-1}_m$ is continuously differentiable. Define $\F_{m+1}$ by

$$F_{m+1}(y)=B_m \circ F_m \circ G_m^{-1} (y), \qquad y \in V_{m+1} \tag{5}$$

Then $\F_{m+1}\in C^1(V_{m+1}),\F_{m+1}(\0)=\0$ and $(\d\F_{m+1})_\0$ is invertible by the chain rule. Also, for $\x\in U_m$, $$ \begin{aligned} P_m\F_{m+1}(\G_m(\x)) & = P_m B_m \circ \F_m(\x) \\ & = P_m[P_{m+1}\x+\alpha_k(\x)\mathbf{e}_m+\cdots] \\ & =P_{m-1}\x+ \alpha_k(\x)\mathbf{e}_m \\ &=P_m \G_m (\x) \end{aligned} $$ so that $$ P_m\F_{m+1}(\y)=P_m\y, \qquad \y \in V_{m+1}\,. $$ Take $\F_1=\F$; it follows from the argument above that Equation 3 also holds for $m = n$. Moreover, since $B_m \circ B_m = I$, Equation 5, with $\y=\G_m(\x)$, is equivalent to $$ \F_m(\x)=B_m\circ \F_{m+1} (\G_m(\x)), \qquad \x\in U_m\,. $$ If we apply this with $m = 1, \ldots, n-1$, we successively obtain $$ \begin{aligned} \F=\F_1 & =B_1\circ \F_2 \circ \G_1 \\ &=B_1\circ B_2 \circ \F_3 \circ \G_2 \circ \G_1 = \cdots \\ &=B_1\circ \cdots \circ B_{n-1} \circ \F_n \circ \G_{n-1} \circ \cdots \circ \G_1 \end{aligned} $$ in some neighborhood of $\0$. By Equation 3, $\F_n$ is primitive. This completes the proof.

Partitions of Unity

Suppose that K is a compact subset of $\mathbf{R}^n$ and $\{V_\alpha\}$ is an open cover of K. Then there exist functions $\psi_1,\dots, \psi_s\in C(\R^n)$ such that

(a) $0 \leq \psi_i \leq 1$ for $1 \leq i \leq s$;

(b) each ψ_i has its support in some V_α, and

(c) $\psi_1(\x)+\dots+\psi_s(\x)=1$ for every $\x \in K$.

$\{\psi_i\}$ is called a *partition of unity*, and (b) is sometimes expressed by saying that $\{\psi_i\}$ is subordinate to the cover $\{V_\alpha\}$.

If $f\in C(\R^n)$ and the support of f lies in K, then

$$f = \sum_{i=1}^{s} \psi_i f \,.$$

Each $\psi_i f$ has its support in some V_α.

Change of variables in multiple integrals

Suppose T is a one-to-one C^1-mapping of an open set $E\subset \R^k$ into $\R^k$ such that $J_T(\x)\neq 0$ for all $\x \in E$. If f is a continuous function on $\R^k$ whose support is compact and lies in $T(E)$, then

$$\int_{\R^k} f(\y) \,\d\y= \int_{\R^k} f(T(\x)) \,, |J_T(\x)| \,\d\x. \quad (6)$$

Proof. **step 1**: If T is a primitive mapping, then T only changes one coordinate, say $T(\x)=(x_1, \dots,\hat{x_t},\dots,x_k)$ where $\hat{x_t}=g(\x)$. It follows from the change of variable in integral of one variable that $$ \begin{aligned} \int _{\R^k}f(T(\x)) \, \vert J_T(\x)\vert \d\x & =\int_{\R^k}f(x_1,\dots,\hat{x_t},\dots,x_k)\,\big\vert \frac{\partial g}{\partial x_t}\big\vert\d\x \\ & = \int _{\R^{k-1}}\d\x\int _{\R}f(x_1,\dots,\hat{x_t},\dots,x_k)\big\vert\frac{\partial g}{\partial x_t}\big\vert \d \x_t \\ & = \int _{\R^{k-1}} \d\x \int_{\R} f(x_1,\dots,x_t,\dots,x_n)\d\x_t \\ & = \int _{\R^k} f(\x) \d\x\,. \end{aligned} $$ Thus Equation 6 holds for primitive mappings.

step 2: If T is a flip which interchanges two coordinates, say x_m and x_n, then Equation 6 also holds, for Lemma 2 and $$ \int_{\R^k} f(T(\x)) \, \vert J_T(\x) \vert \d\x = \int _{\R^k} f(T\x)\d\x\,. $$

step 3: If Equation 6 holds for T and Q, then it also holds for $S = T \circ Q$. Because: $$ J_{T\circ Q}(\x)=J_T(\x) \cdot J_Q(\x). $$

step 4: Each point $\mathbf{a} \in E$ has a neighborhood $U \subset E$ in which $$ T(\x)=T(\mathbf{a})+B_1\cdots B_{k-1}\G_k \circ \G_{k-1}\circ \cdots \circ \G_1 (\x -\mathbf{a}), $$ where $\G_i$ and B_i are as in Equation 2. Setting $V = T(U)$, it follows that Equation 6 holds if the support of f lies in V. Thus:

Each point $\y\in T(E)$ lies in an open set $V_{\y}\subset T(E)$ such that Equation 6 holds for all continuous functions whose support lies in $V_{\y}$.

step 5: Now let f be a continuous function with compact support $K \subset T(E)$. Since $\{V_{\y}\}$ covers K, the *Corollary of partition of unity* shows that $f = \sum \psi_i f$, where each ψ_i is continuous and each ψ_i has its support in some $V_{\y}$. Thus Equation 6 holds for each $\psi_i f$, and hence also for their sum f.